## 3.3 Flash sync and intensity

Typical high-end flash for a digital SLR camera

Practically, all digital cameras include a built-in electronic flash (also called a strobe light). A flash has two principal purposes: (a) to light up a subject when there's not enough available light, and (b) filling up light in the dark areas and shadowy regions of the scene (flash fill). In short, the flash adds to the intensity of the available light or provides an even lighting to ensure a consistent exposure to all elements of the photographic image.

Here's a list of the typical flash modes usually found on digital cameras.

Auto: The camera decides when the flash is required and fires automatically

Manual: Gets the camera to flash even if there's plenty of light around, using the flash as a fill-in to eliminate the intrusive shadows that could mar your photograph.

Slow Sync: The camera judges exposure by measuring the light from the flash, and not from elements in the background. This ensures that the exposure occurs as if flash was turned off, and then at the end of the exposure, before the shutter closes, the flash is fired to light up the foreground subject.

Red-eye: 'Red-eye' is a phenomenon which occurs when light bounces off the retina of the eye of a subject and cause the eyes to appear red. The 'red-eye' flash mode reduces the red-eye by either firing the flash twice or thrice in quick succession before the actual shot or by turning on a small white light on the camera. The first method is far more effective.

No flash: Disables the flash if you don't think it's going to help. This is good for shots with minimal light (sunsets, city life at night, etc.).

### How the flash works

Earlier, the badly needed flash of light was provided by ignit-